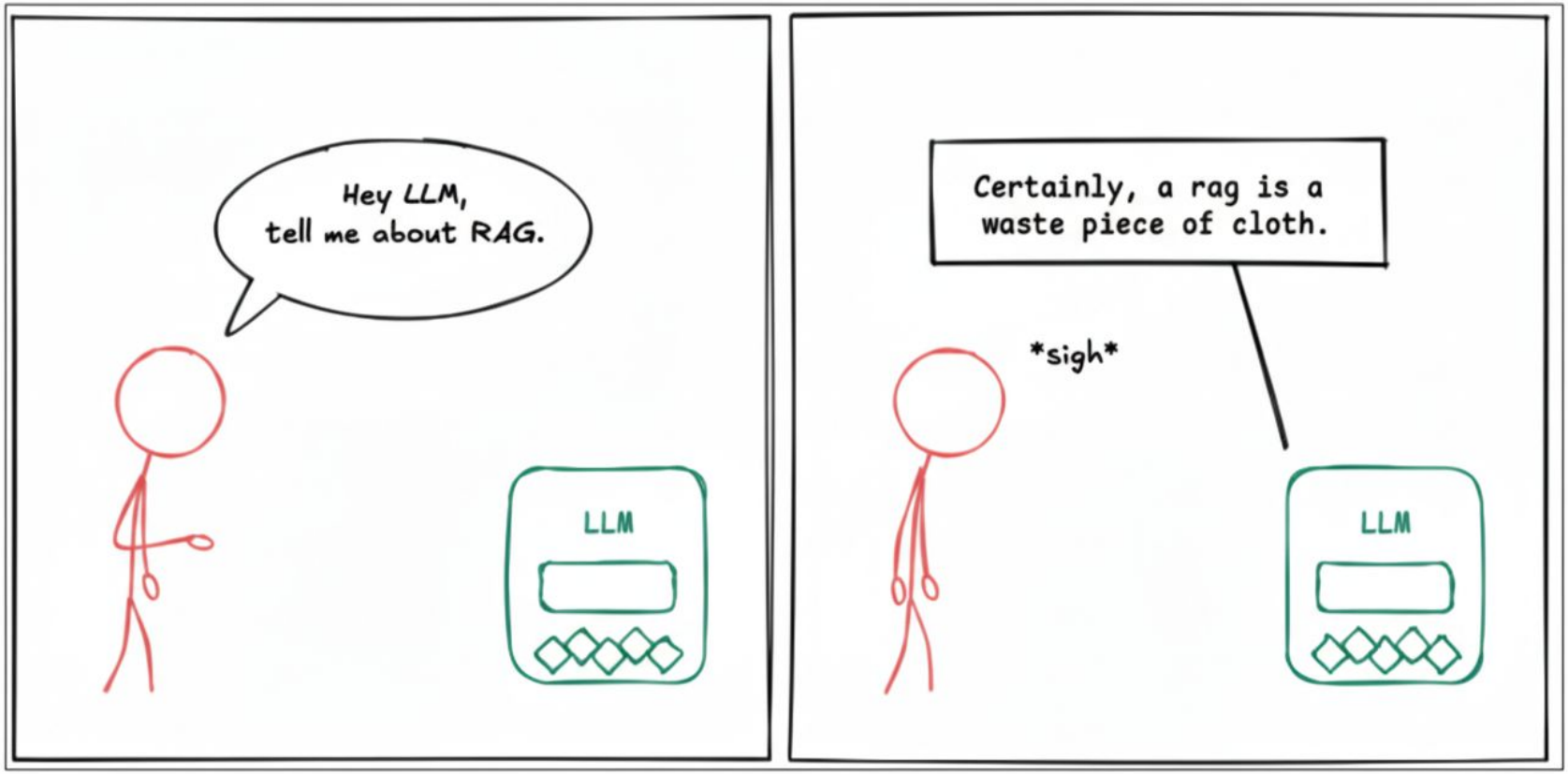
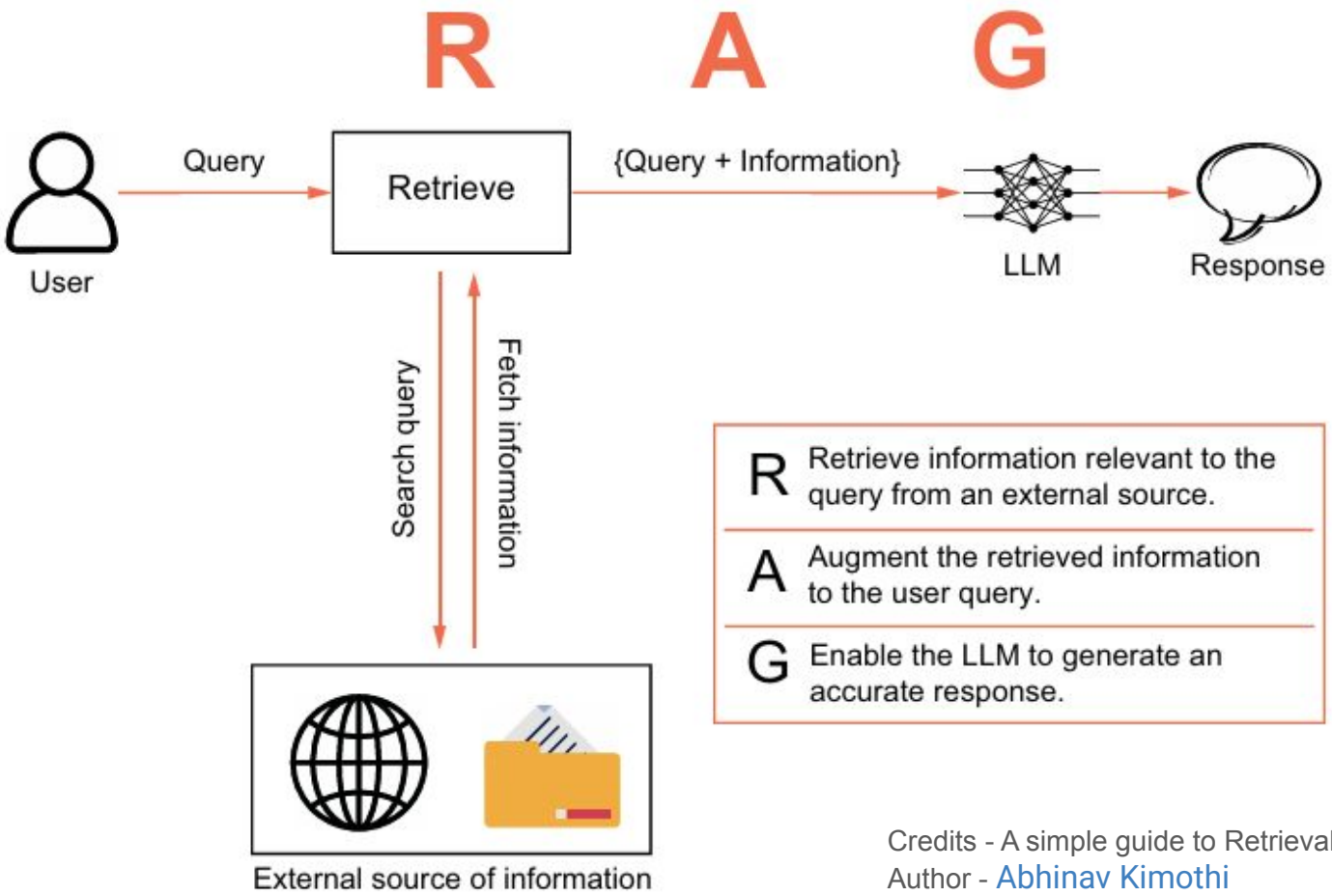

Hey LLM,
tell me about RAG.
LLM
Certainly, a rag is a
waste piece of cloth.
sigh
LLM

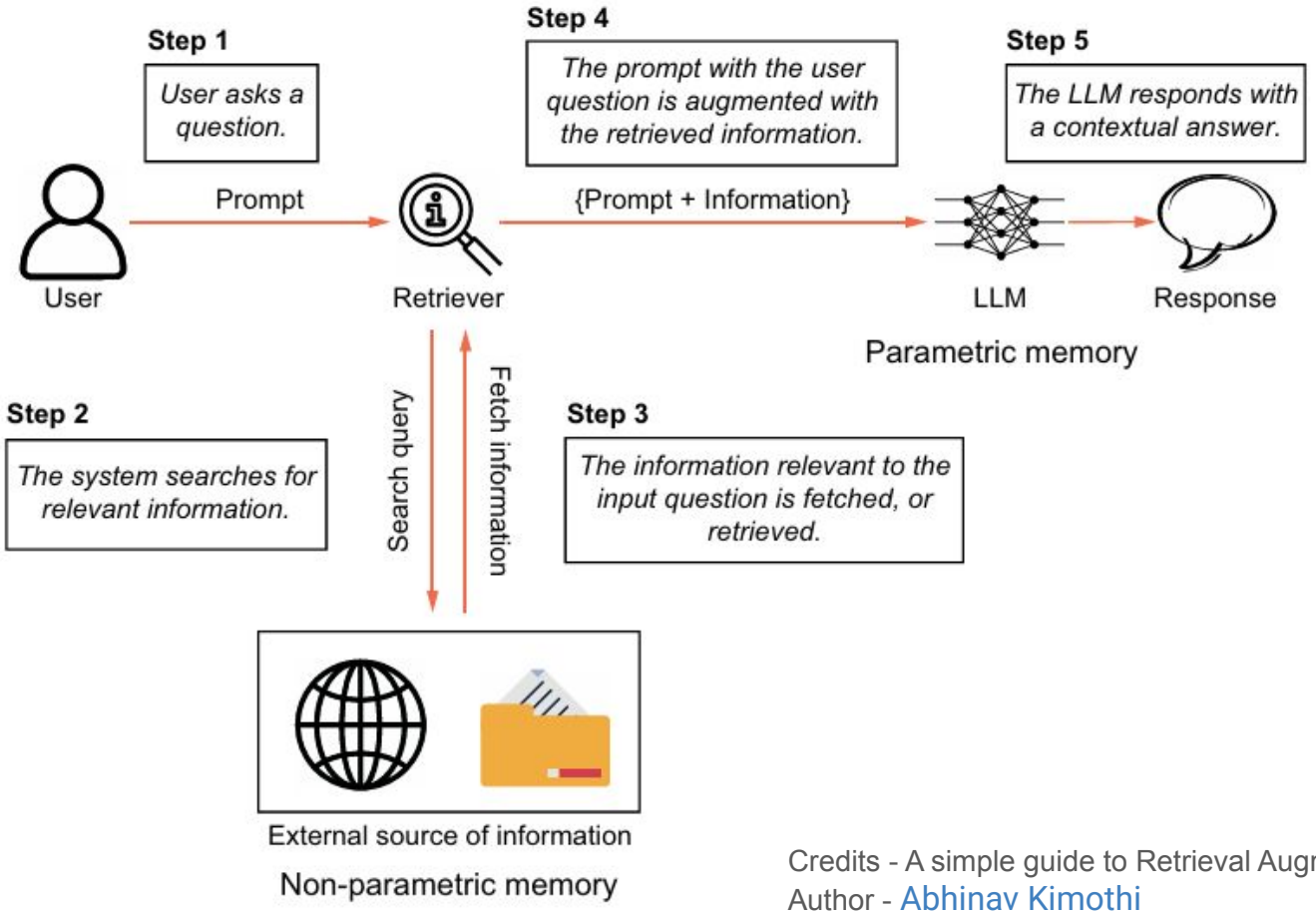
Overview of RAG



R	Retrieve information relevant to the query from an external source.
A	Augment the retrieved information to the user query.
G	Enable the LLM to generate an accurate response.

Credits - A simple guide to Retrieval Augmented Generation
Author - Abhinav Kimothi

Parametric Memory & Non Parametric Memory



Credits - A simple guide to Retrieval Augmented Generation
Author - Abhinav Kimothi

Indexing Pipeline

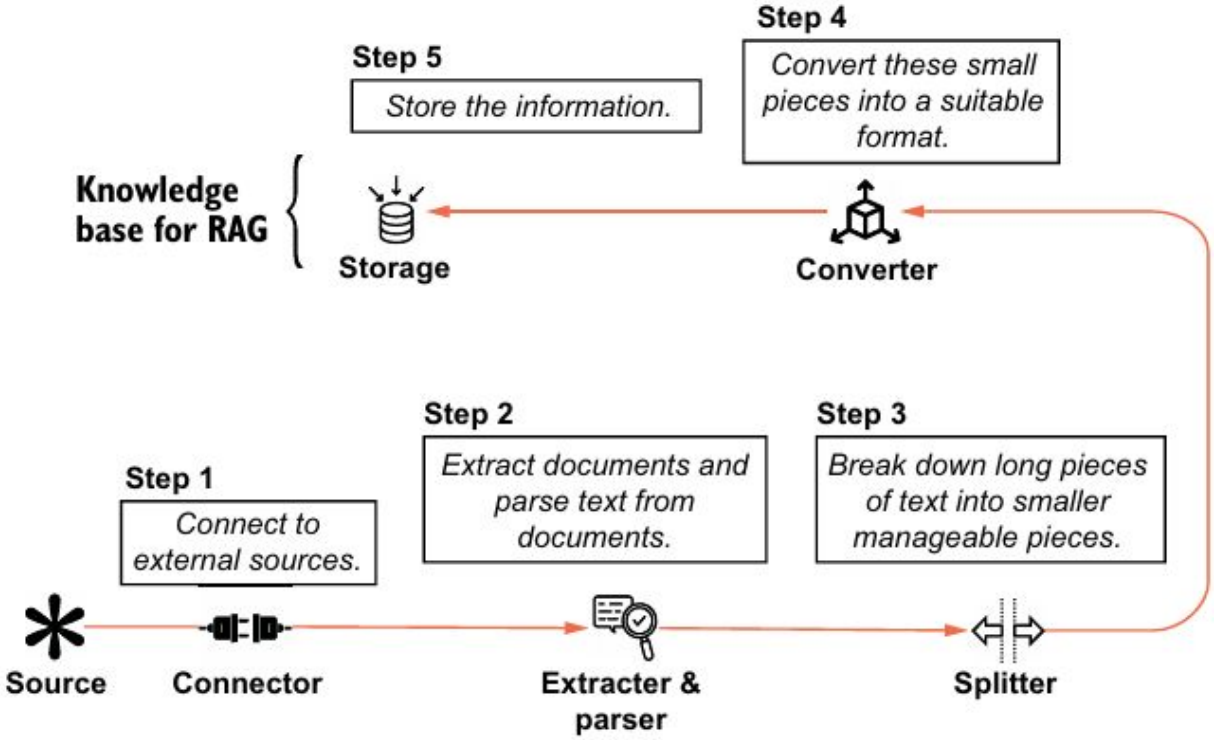


Figure 2.2 Indexing pipeline covering the steps to create the knowledge base for RAG. This involves connecting to the source, parsing, splitting, converting, and storing information.

Credits - A simple guide to Retrieval Augmented Generation
Author - Abhinav Kimothi

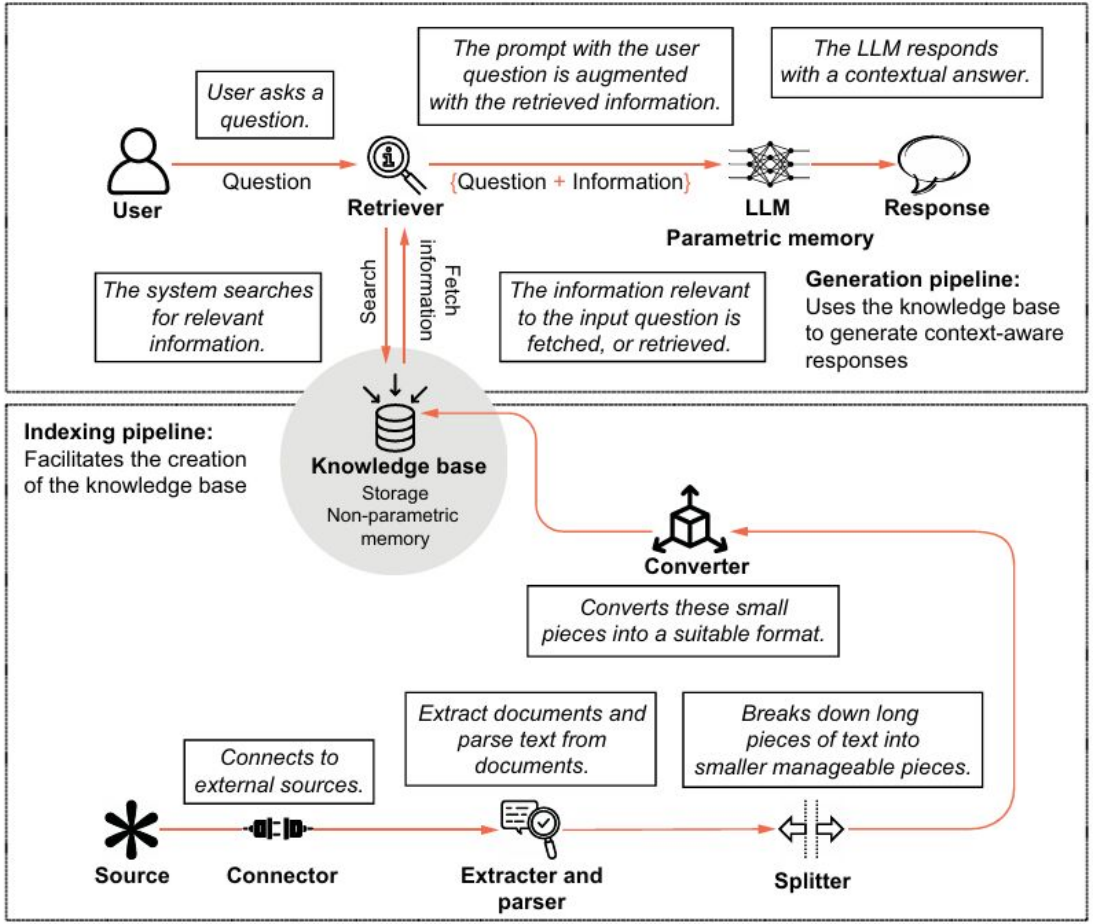


Figure 2.3 The indexing and generation pipelines together make a RAG system. The indexing pipeline is an offline process, while the generation pipeline facilitates real-time interaction with the knowledge base.

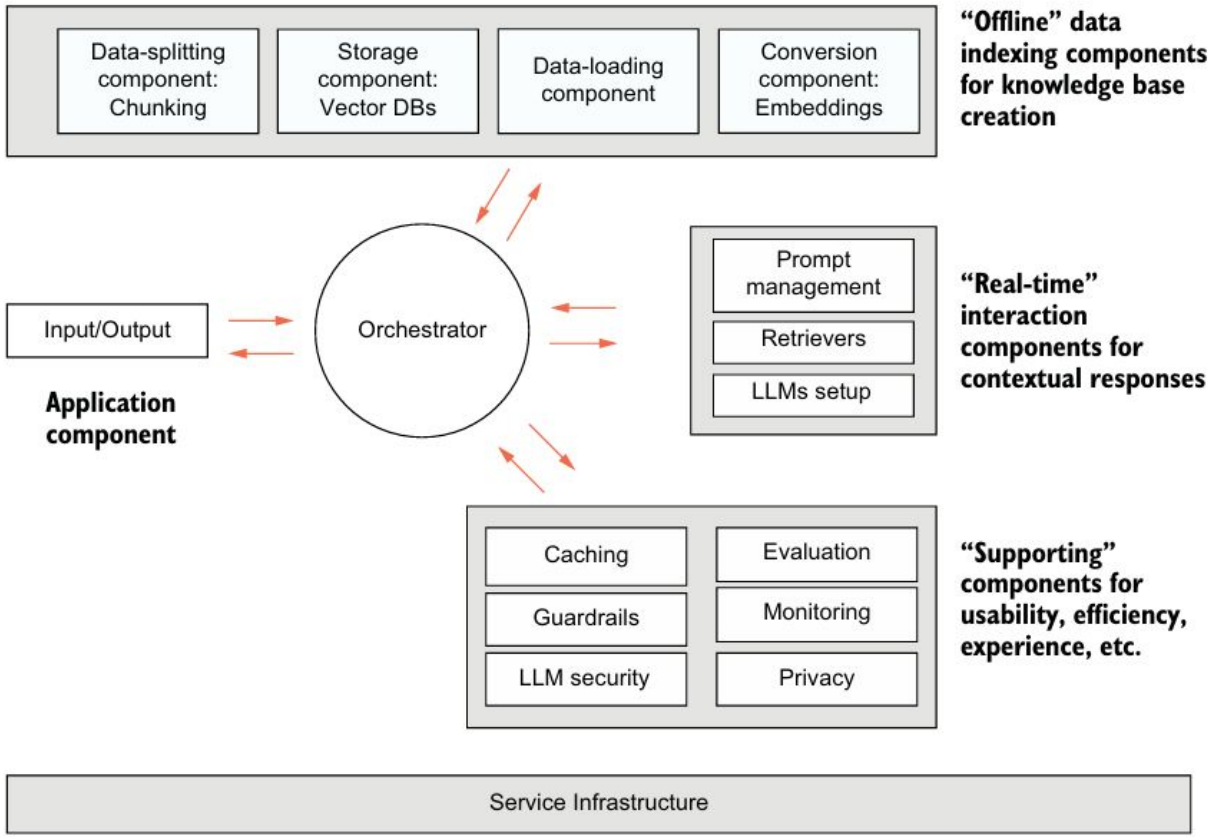


Figure 2.4 Components of a production-ready RAG system

Credits - A simple guide to Retrieval Augmented Generation
Author - Abhinav Kimothi

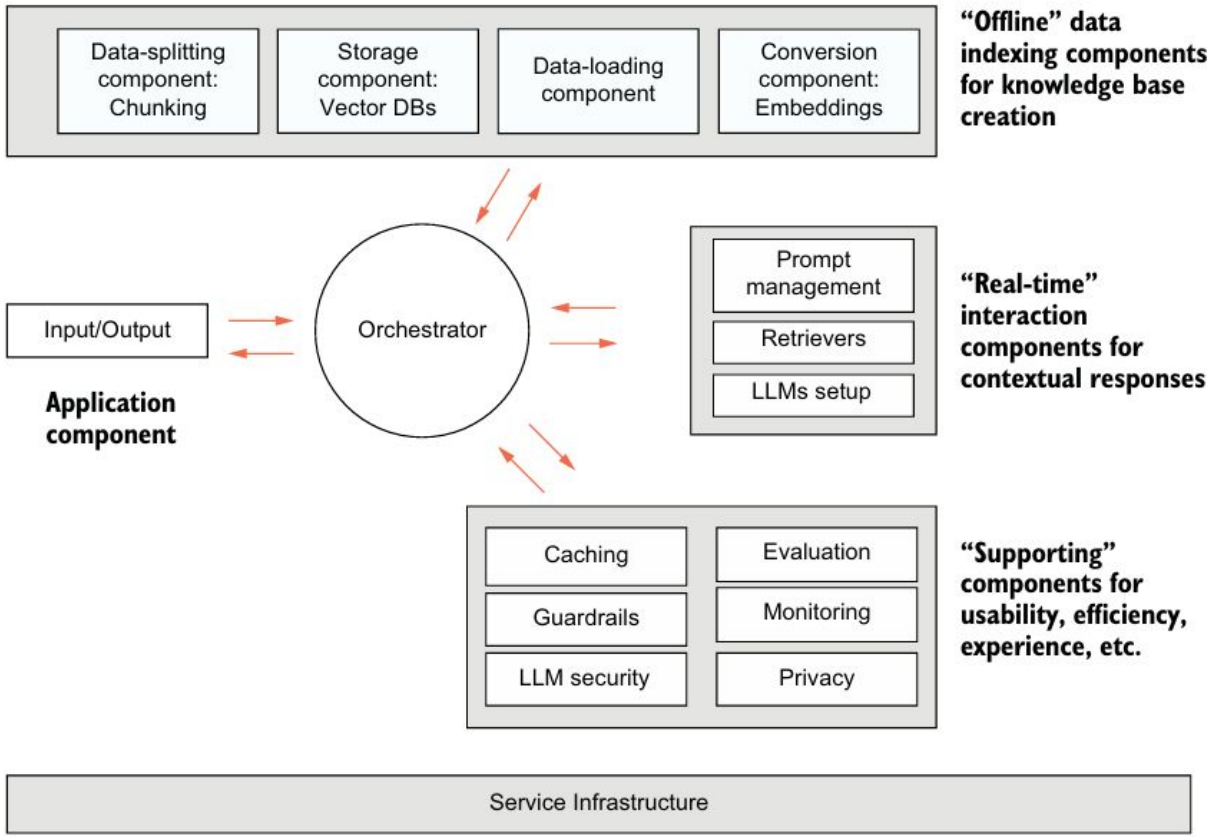
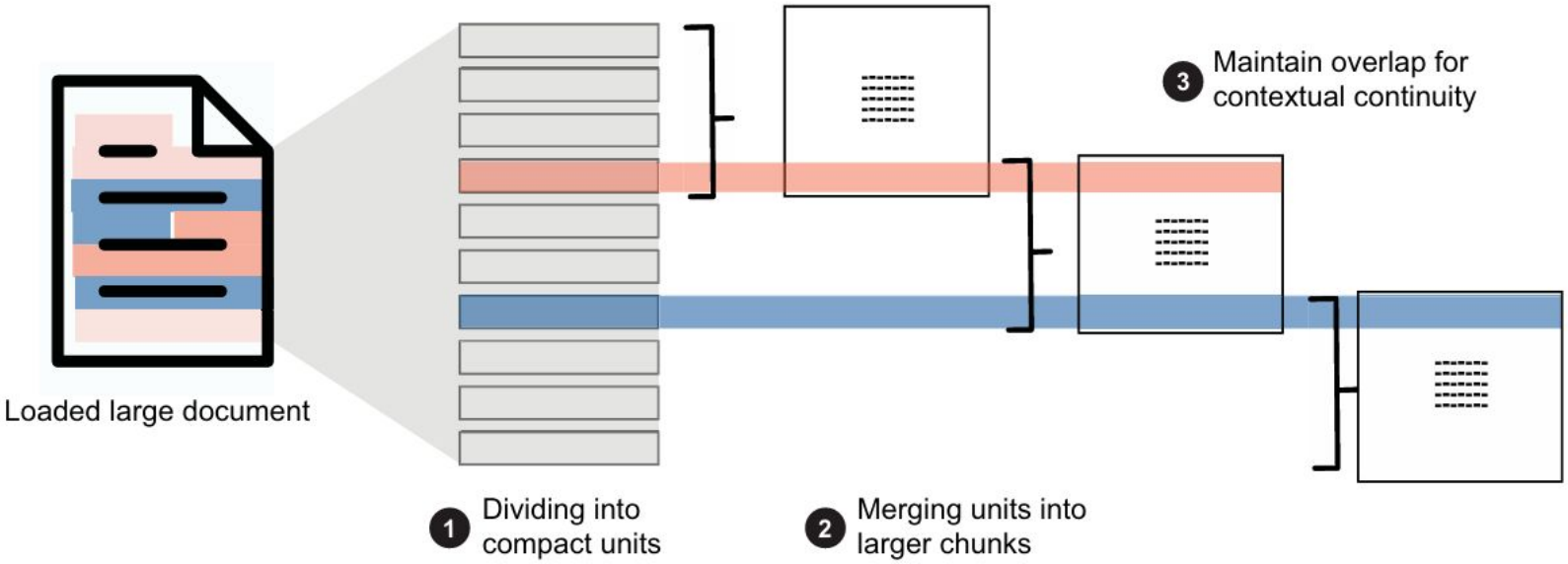


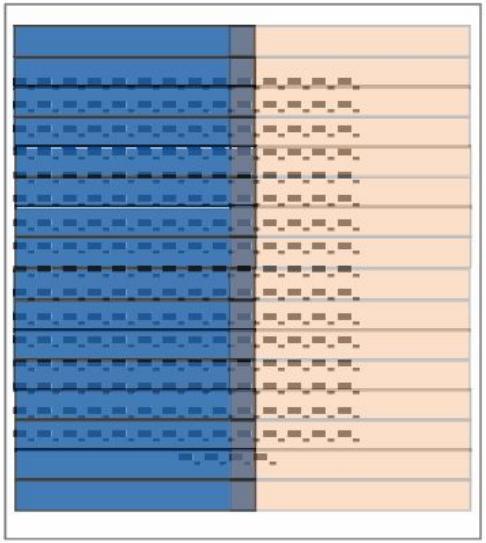
Figure 2.4 Components of a production-ready RAG system

Credits - A simple guide to Retrieval Augmented Generation
Author - Abhinav Kimothi



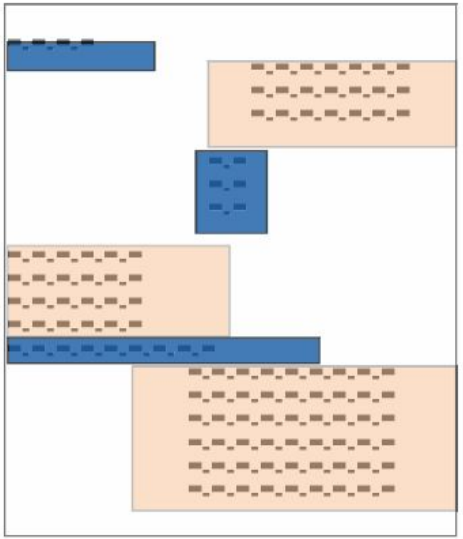
Small to Big Chunking

Credits - A simple guide to Retrieval Augmented Generation
Author - Abhinav Kimothi



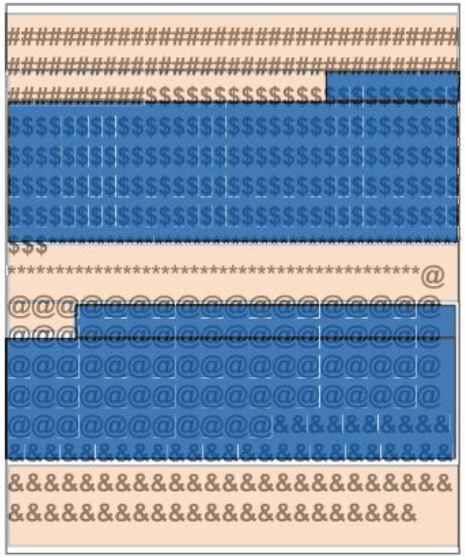
Fixed-size chunking

Te length of chunks is uniform and predetermined with some overlap.



Specialized chunking

Chunking is carried out depending on the structure of the input.



Semantic chunking

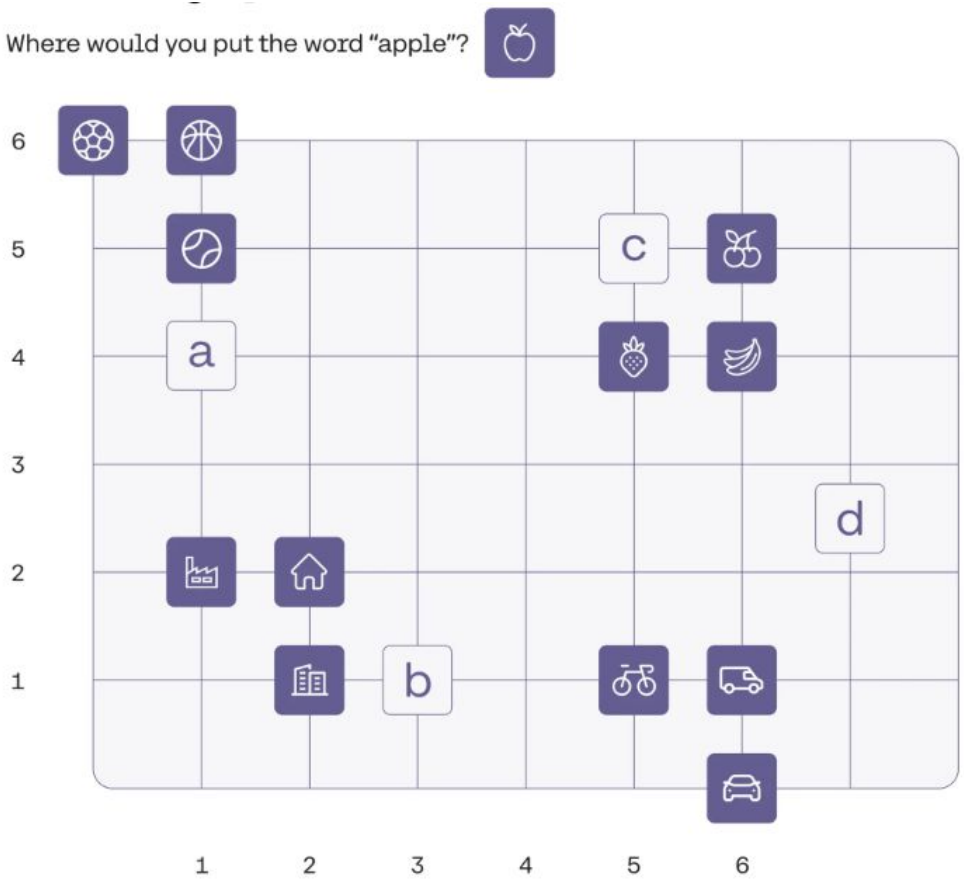
Chunking preserves the semantic integrity or the meaning of the input.

Figure 3.3 Chunking methods

Credits - A simple guide to Retrieval Augmented Generation
Author - Abhinav Kimothi

A small Quiz !

Banana
Basketball
Bicycle
Building
Car
Castle
Cherry
House
Soccer
Strawberry
Tennis
Truck

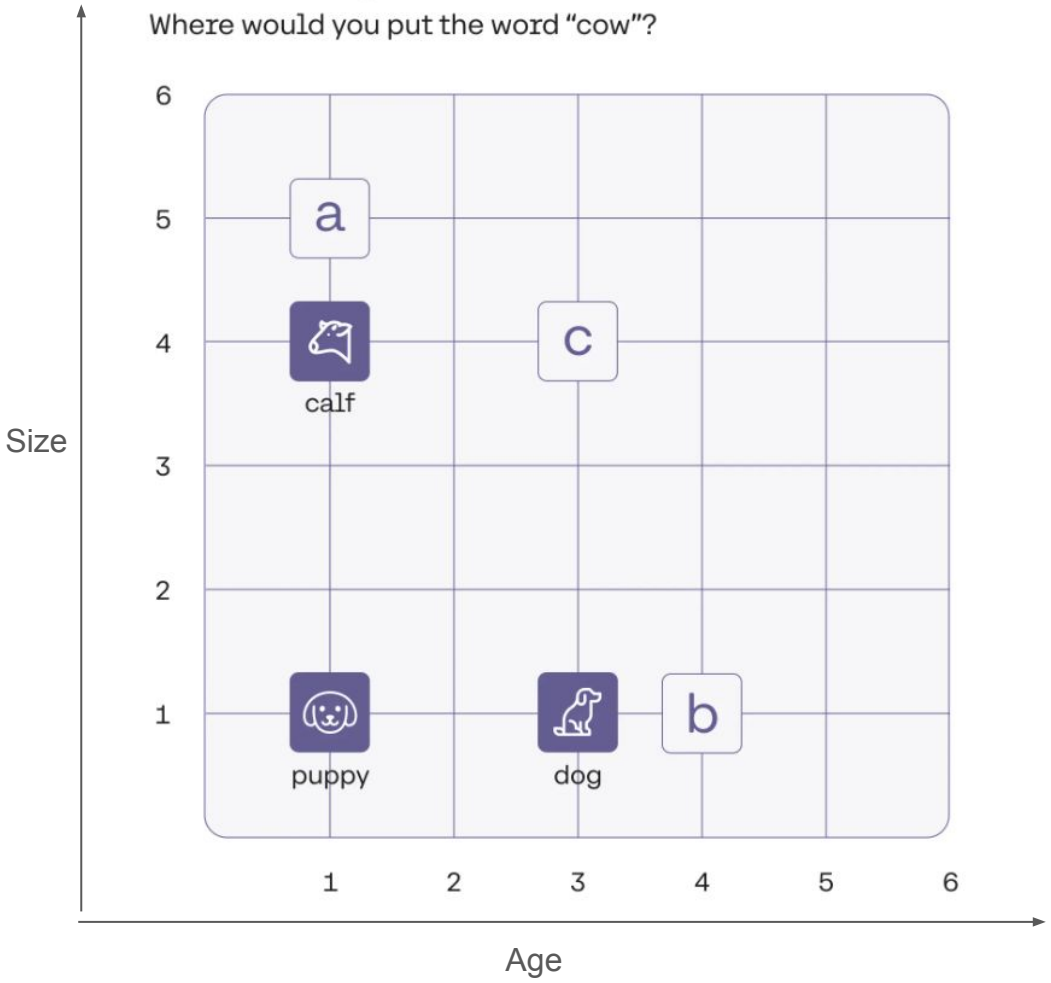


Credits - [Cohere blog]

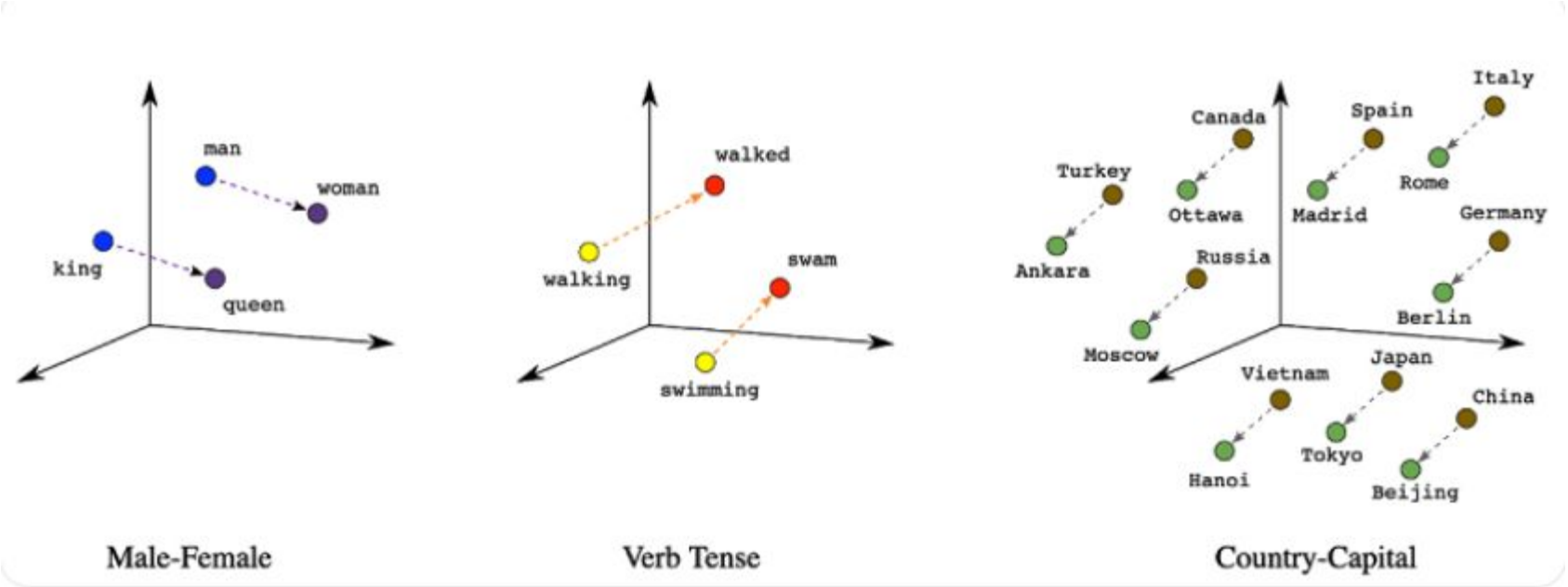
Puppy
Cow
Dog
Calf

Where do we add the word Cow ?

Where do we add the word Whale ?



Three Dimensional Space Example



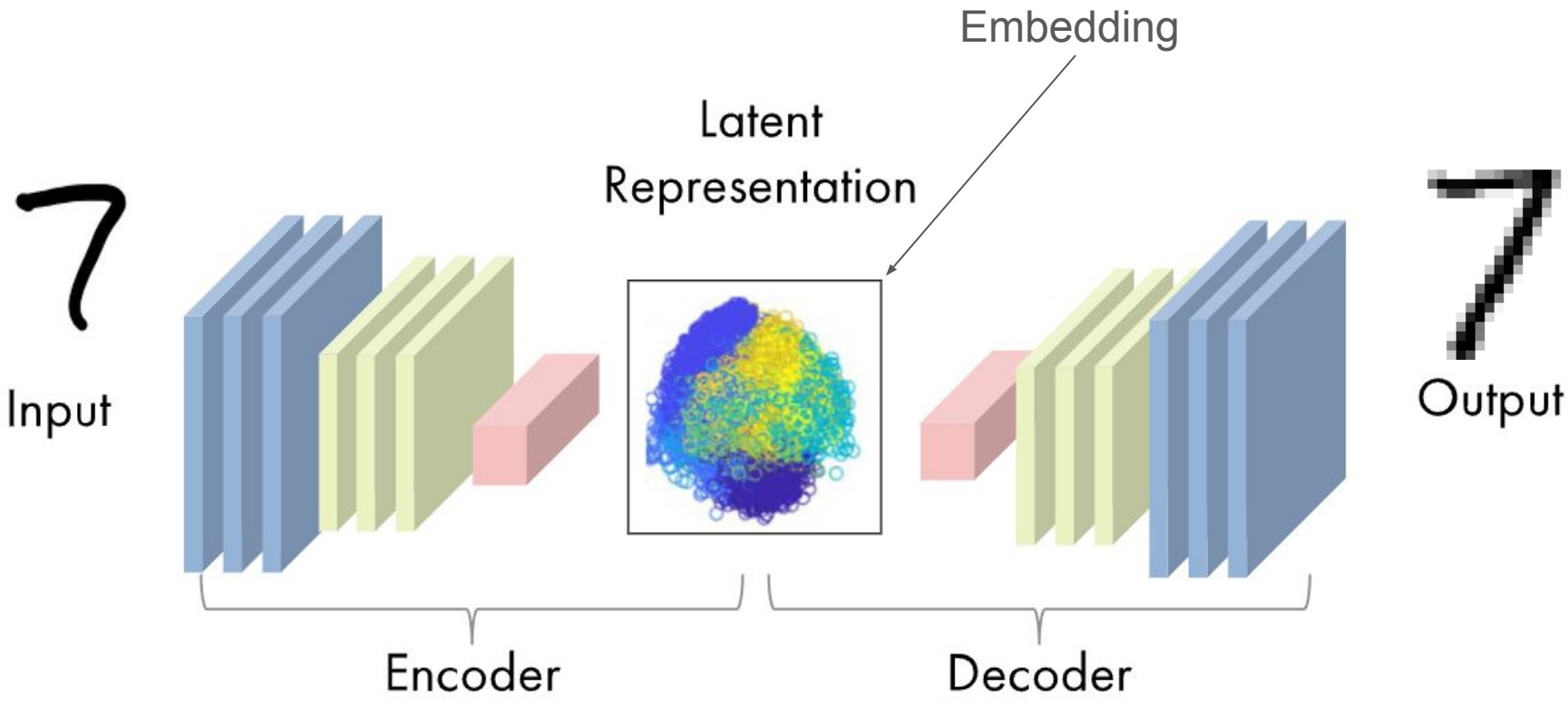


Figure 1: An autoencoder consists of an encoder and a decoder.

Sentence Embedding Models

USE (Universal Sentence Encoder) by Google

SBERT (Sentence Bert) - Nils Rimers and Iryna Gurevych

DPR (dense passage retrieval) by facebook and others

Sentence-T5: a variant of the famous T5 model specifically trained for sentence embedding

E5: Using signals from curated multi-domain datasets to achieve SOTA sentence embedding model performance.

DRAGON: A technique that uses large scale synthetic data and curriculum learning to improve retrieval performance with sentence embedding models

COLBERT: Using multiple vectors to represent sentence embeddings and "late" interaction

Pure Sentence Similarity vs. Question Answering

- **Pure Similarity**: do sentences A and B have similar semantic meaning?
- **Question Answering**:
  - Given question Q

    and dataset A(1), A(2), ..., A(n)
  - Rank the possible responses A(i) by relevance to Q

Are these the same?

Example: pure similarity vs Q/A

Our dataset includes 4 possible responses:

- A1: “What is the tallest mountain in the world?”
- A2: “Mount Everest is the tallest.”
- A3: “Mount Shasta is very tall”
- A4: “I want to ride a bike”

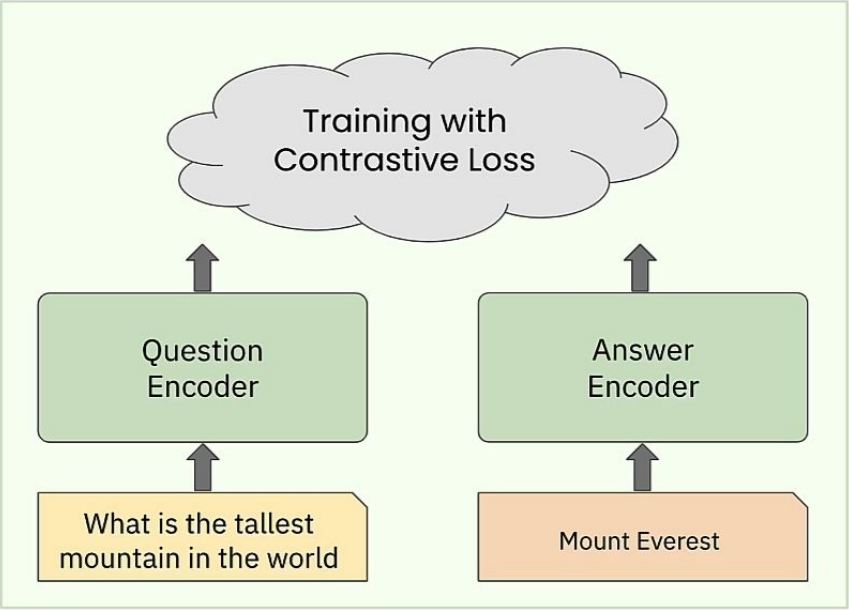
Question: “What is the tallest mountain in the world?”

↓

Which one do we want to rank the highest?

The Dual Encoder Architecture

- Separate question/answer encoders
- Trained with contrastive loss

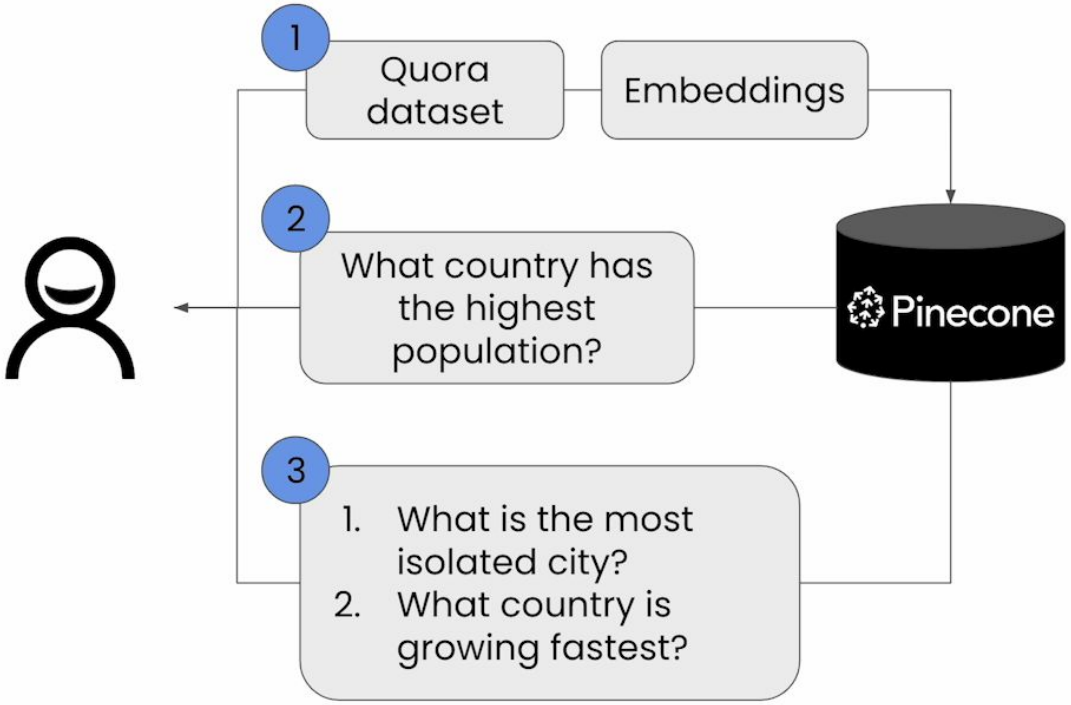


Let’s try it out

https://dashboard.cohere.com/playground/embed

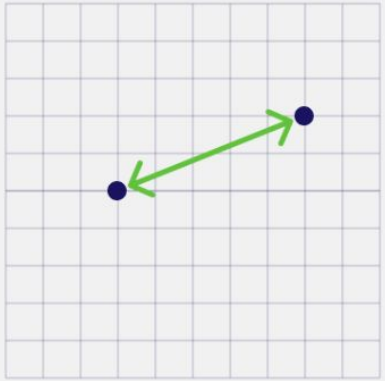
Applications of Vector Database

Semantic Search



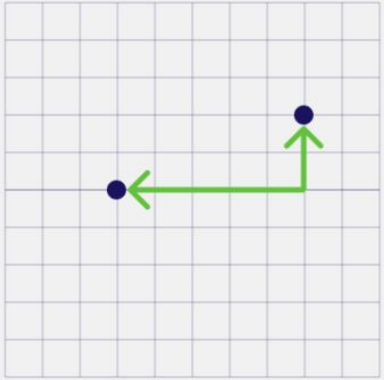
Let’s Visualise it using Atlas Embeddings

https://apple.github.io/embedding-atlas/



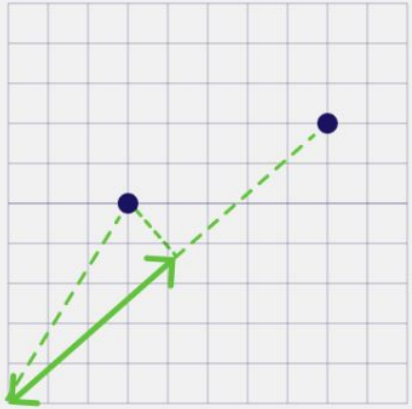
Squared Euclidean (L2 Squared)

$$\sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$



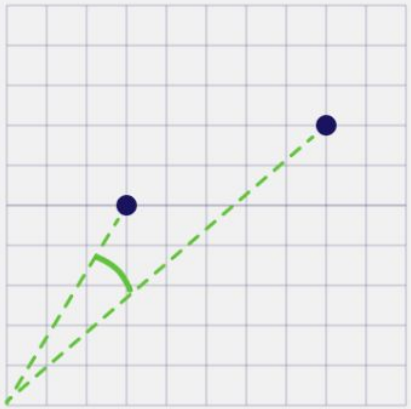
Manhattan (L1)

$$\sum_{i=1}^{n} |x_i - y_i|$$



Dot Product

$$A \cdot B = \sum_{i=1}^{n} A_i B_i$$



Cosine Distance

$$1 - \frac{A \cdot B}{||A|| \quad ||B||}$$